Combined vertices in the pure-recoil limit

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Abstract

We isolate the combined vertex $V/\sqrt{C}$ in the internal integral representation of the continued tetrahedron graph. An exact cancellation, followed by two inverse-b shifts, gives its normalized pure-recoil limit as an ordinary Gamma quotient. The doubled vertex has a simple hyperbolic form. Away from mass-triangle degeneration, four such vertices reduce to a mass polynomial, uniformly on a specified transverse box. This does not identify the physical four-point normalization or complete the internal integral. In particular this vertex must not be multiplied onto a full-6j Gram asymptotic without redoing the prefactor bookkeeping. All results are author self-reviewed drafts, without a priority claim.

1 Which factors are being computed

Set the de Sitter radius to one. In CLP [1], (3.17) and (3.43)–(3.47), write $Q = b + b^{-1}$, $p_a = iP_a$, and $x_\sigma = Q/2 + \sum_{a=1}^{3} \sigma_a p_a$. Let $\mathcal{E} = \{\sigma : \sigma_1\sigma_2\sigma_3 = +1\}$ and $\mathcal{O} = -\mathcal{E}$. Then

$$C = K_b \frac{\prod_{\sigma\in\mathcal{E}\cup\mathcal{O}} \Gamma_b(x_\sigma)}{\prod_a \Gamma_b(Q+2p_a)\Gamma_b(Q-2p_a)}, \qquad K_b = \frac{\Gamma_b(2Q)}{\sqrt{2}\,\Gamma_b(Q)^3}, \tag{1}$$

$$V^2 = \prod_{\sigma\in\mathcal{E}} S_b(x_\sigma), \qquad S_b(x) = \frac{\Gamma_b(x)}{\Gamma_b(Q-x)}. \tag{2}$$

Consequently, as a meromorphic identity,

$$U_b := \frac{V^2}{C} = K_b^{-1} \frac{\prod_a \Gamma_b(Q+2p_a)\Gamma_b(Q-2p_a)}{\prod_{\sigma\in\mathcal{O}} \Gamma_b(x_\sigma)^2}. \tag{3}$$

Square-root branches belong to the original continued graph. On the real Euclidean germ the doubled combined vertex is $T_b = |V/\sqrt{C}|^2 = |U_b|$.

The whole graph can thus be written as

$$Z_{\text{tet}} = \left(\prod_{\text{four vertices}} T_b\right) |\mathcal{J}_b|^2, \qquad \{6j\}_b = \left(\prod_{\text{four vertices}} V_b\right) \mathcal{J}_b. \tag{4}$$

Here $\mathcal{J}_b$ is precisely the internal double-sine integral in CLP (3.46), with its specified contour. This rearrangement does not supply physical insertion legs or external energy cycles.

2 A normalized pure-recoil vertex

Use $b^2 = i/\kappa$ and the pure-recoil parametrization

$$p_a = -\frac{1}{2b} - \frac{ibm_a}{2}, \quad m_a > 0, \qquad L = b^{-2} = -i\kappa, \quad S = m_1 + m_2 + m_3.$$

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Define

$$z_0 = \frac{1+iS}{2}, \qquad z_a = \frac{1-i(S-2m_a)}{2} \quad (a=1,2,3).$$

The four odd arguments in (3) are $2/b + bz_0, bz_1, bz_2, bz_3$. The six leg arguments are $b(1 - im_a), 2/b + b(1 + im_a)$. The exact shift relation gives

$$\Gamma_b(2/b + bz) = \frac{2\pi\, b^{1-L-2z}}{\Gamma(z)\Gamma(L+z)}\Gamma_b(bz). \tag{5}$$

All powers of b generated in (3) have mass-independent net exponent. Up to a factor depending only on b, one obtains

$$U_b(\mathbf{m}) = E_b(\mathbf{m})\frac{\Gamma(z_0)^2\Gamma(L+z_0)^2}{\prod_a \Gamma(1+im_a)\Gamma(L+1+im_a)}, \qquad E_b = \frac{\prod_{a,\pm}\Gamma_b(b(1\pm im_a))}{\prod_{j=0}^3 \Gamma_b(bz_j)^2}. \tag{6}$$

Proposition 1. *For fixed positive real masses* $\mathbf{m}$ *and a fixed positive reference triple* $\mathbf{r}$*, with the branches inherited from CLP,*

$$\lim_{\kappa\to\infty}\frac{U_b(\mathbf{m})}{U_b(\mathbf{r})} = \frac{F(\mathbf{m})}{F(\mathbf{r})}, \qquad F(\mathbf{m}) = \frac{\prod_a \Gamma(1-im_a)}{\prod_a \Gamma(\frac{1}{2} - \frac{i}{2}(S-2m_a))^2}. \tag{7}$$

In particular $T_b(\mathbf{m})/T_b(\mathbf{r}) \to T(\mathbf{m})/T(\mathbf{r})$*, where*

$$T(\mathbf{m}) = \frac{1}{\pi^{3/2}}\sqrt{\frac{m_1m_2m_3}{\prod_a \sinh(\pi m_a)}}\prod_a \cosh\left(\frac{\pi(S-2m_a)}{2}\right). \tag{8}$$

Only a mass-independent normalization has been suppressed.

Proof. The usual small-b reduction to Gamma functions is familiar from the Schwarzian limit [2]. To control this particular balanced quotient on $b^2 = i/\kappa$, a direct argument avoids assuming a uniform continuation of that real-b formula.

Put $f_\kappa(z) = \log\Gamma_b(bz) - \log\Gamma(z)$. The third logarithmic derivative of the canonical Barnes product is absolutely convergent and gives, on the pole-free domain in use,

$$f_\kappa'''(z) = -2\sum_{q=1}^{\infty}\sum_{n=0}^{\infty}(z+n-iq\kappa)^{-3}. \tag{9}$$

This also follows by differentiating the Barnes double-zeta definition three times; quadratic normalization terms disappear. For $\operatorname{Re} z \geq 1/2$, $|\operatorname{Im} z| \leq R$, and $\kappa \geq 2R$, comparison of the n-sum with an integral gives

$$|f_\kappa'''(z)| \leq K_\kappa := \frac{16\zeta(3)}{\kappa^3} + \frac{8\zeta(2)}{\kappa^2}. \tag{10}$$

Indeed, with $A = q\kappa/2$, the sum of $((n+1/2)^2 + A^2)^{-3/2}$ is at most $A^{-3} + A^{-2}$.

The signed arguments in E_b, with coefficients $+1$ for each $1 \pm im_a$ and -2 for each z_j, satisfy

$$\sum c = -2, \qquad \sum cz = 2, \qquad \sum cz^2 = 4. \tag{11}$$

All three are independent of the masses. Taylor-expand f_κ about $z = 1$ along line segments in the domain above. The quadratic polynomial cancels between a triple and its reference. The remaining logarithmic error is at most

$$\frac{K_\kappa}{6}\left(\sum_{\mathbf{m}}|c|\,|z-1|^3 + \sum_{\mathbf{r}}|c|\,|z-1|^3\right). \tag{12}$$

Thus $E_b(\mathbf{m})/E_b(\mathbf{r})$ has the corresponding ordinary Gamma quotient as its limit. This establishes the needed normalized reduction on this ray without choosing an absolute Barnes normalization.

Finally put $B_L(\mathbf{m}) = \Gamma(L+z_0)^2/\prod_a \Gamma(L+1+im_a)$. The fixed-shift Gamma ratio asymptotics [3] give

$$B_L(\mathbf{m}) \sim \Gamma(L)^{-1} L^{-2},$$

since $2z_0 - \sum_a (1+im_a) = -2$. Its ratio to the reference tends to one. Substitution into (6) proves (7). The identities $|\Gamma(1-im)|^2 = \pi m/\sinh(\pi m)$ and $|\Gamma(1/2-ix/2)|^2 = \pi/\cosh(\pi x/2)$ give (8). □

Equation (12) controls the balanced Barnes part only. The fixed-shift estimate for B_L has not here been replaced by a fully quantified joint growing-mass bound. We therefore use the sequential limit: first $\kappa \to \infty$ at fixed masses, then the large-mass regime below. No finite-G error estimate follows.

3 A local transverse box, with an explicit vertex remainder

For real positive masses satisfying strict triangle inequalities, set $\ell_a = S - 2m_a > 0$. Equation (8) factors exactly as

$$T(\mathbf{m}) = \frac{\sqrt{m_1 m_2 m_3}}{2\sqrt{2}\,\pi^{3/2}} \mathcal{E}(\mathbf{m}), \qquad \mathcal{E} = \frac{\prod_a (1+e^{-\pi \ell_a})}{\prod_a (1-e^{-2\pi m_a})^{1/2}}. \tag{13}$$

The exponentials cancel because $\sum_a \ell_a = S$. Continue this expression from that real germ. In a domain with $\operatorname{Re}\ell_a \geq d > 0$ and $\operatorname{Re} m_a \geq a > 0$, the square root and logarithm fixed there obey

$$|\log \mathcal{E}| \leq \frac{3e^{-\pi d}}{1-e^{-\pi d}} + \frac{3e^{-2\pi a}}{2(1-e^{-2\pi a})}. \tag{14}$$

This is an analytic continuation, not a new pointwise modulus at complex masses.

For equal probes ν, write the cyclic observer masses

$$m_1 = M + \omega/2 + u, \quad m_3 = M + \omega/2 - u, \qquad m_2 = M - \omega/2 + v, \quad m_4 = M - \omega/2 - v.$$

The four vertex triples are (m_i, m_{i+1}, ν), cyclically. Every graph edge occurs twice. Their product, normalized at $\omega = u = v = 0$, is therefore

$$\mathcal{R}_V = P \, \exp\{\Delta \log \mathcal{E}_{\text{four}}\}, \tag{15}$$

$$P = \frac{[(M+\omega/2)^2 - u^2][(M-\omega/2)^2 - v^2]}{M^4}. \tag{16}$$

The logarithmic remainder is bounded by the sum of eight bounds (14), four at each configuration.

Take $\omega = iA_0 y$, $A_0 = \nu^2/(2M)$, $|y| \leq Y$, and real $|u|, |v| \leq L_{\text{box}}\sqrt{M}$. If

$$\nu/M \to 0, \qquad L_{\text{box}}\sqrt{M} = o(\nu), \qquad \nu \to \infty,$$

then all the real parts of the triangle gaps are at least $\nu - 2L_{\text{box}}\sqrt{M}$, except for the much larger gaps $m_i + m_{i+1} - \nu$. The exponentials are uniformly small. Moreover, with $a = \omega/(2M), r = u/M, s = v/M$,

$$P - 1 = -2a^2 + a^4 - (r^2+s^2)(1+a^2) + 2a(r^2-s^2) + r^2 s^2.$$

It follows uniformly on this box that

$$\mathcal{R}_V = 1 + O_Y\left(\frac{\nu^4}{M^4} + \frac{L_{\text{box}}^2}{M}\right) + O\left(e^{-\pi(\nu - 2L_{\text{box}}\sqrt{M})}\right). \tag{17}$$

For example the earlier hierarchy $\nu^{4/3} \ll M \ll \nu^2$ allows such boxes, with a slowly growing L_{box}. This controls the limiting vertex on a local box; it does not bound the full integration tail or prove the action's Gaussian approximation.

4 Known observer factors and the remaining physical gap

CLP (3.10) gives the observer factor $Z_{\rm obs}(\alpha) = 32\sin^2(\alpha/2)\sinh^2(\kappa\alpha/2)$. For $\alpha_i = 2\pi(1 - m_i/\kappa)$ and $\sum_i m_i = 4M$, it implies

$$\lim_{\kappa\to\infty} \frac{\prod_{i=1}^4 Z_{\rm obs}(\alpha_i)}{Z_{\rm obs}(2\pi(1 - M/\kappa))^4} = P^2. \tag{18}$$

The common entropy and Boltzmann exponent cancel at fixed total mass. This is a check of a known building block, not a prescription that the physical four-point function contains precisely this product.

The two-point insertion factor in CLP (3.37) is only a leading fixed-probe prescription. Its displayed $(\alpha_1\alpha_2)^{-3/2}$ dependence becomes constant in the sequential pure-recoil limit, but its unspecified correction and the four-point insertion rule are not determined by this observation. We do not impose them.

There is also a bookkeeping correction to guard against. The previous Gram/transverse calculation used a *full-6j* geometric one-loop hypothesis. The V's in (4) are already part of that $6j$. Hence $\mathcal{R}_V$ is not a new factor to multiply into the previous Gram answer. It tells us that in the *internal-integral representation* the combined vertices are locally flat after the specified limits. The determinant and order-one factor of $\mathcal{J}_b$ must be computed in that same representation before the two organizations can be compared.

The next step is therefore the pure-recoil degeneration of the full internal double-sine integral, retaining u, v, followed by its contour and saddle analysis. The physical external-energy cycle, normalization, possible additional saddles and finite-κ uniformity remain independent requirements. No full OTOC, microscopic dictionary, or de Sitter dual has been established.

5 Reproducibility

The script `code/recoil_vertex.py` records 42 checks at 65 decimal digits: three exact signed-moment identities, four Gamma/hyperbolic identities and their remainder bounds, six large-Gamma shift ratios, four lattice-derivative bounds, nine complex transverse-box factorizations and bounds, and three observer ratios. The six shift ratios and three observer ratios are repeated at 95 digits. For the lattice derivative the n-sum is a polygamma function and the q-sum is truncated at 40 with the explicit absolute tail bound from (10). No direct numerical evaluation of the entire double-Gamma quotient or $6j$ is claimed. Finite samples do not certify the uniform statements or a physical finite-G amplitude.

References

[1] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1, (3.10), (3.17), (3.35)–(3.47) and Appendix A.

[2] T. G. Mertens, G. J. Turiaci and H. L. Verlinde, *Solving the Schwarzian via the conformal bootstrap*, JHEP 08 (2017) 136, Appendix B.1. Online reading only; no new local archive.

[3] NIST Digital Library of Mathematical Functions, Section 5.11, fixed-shift Gamma asymptotics.